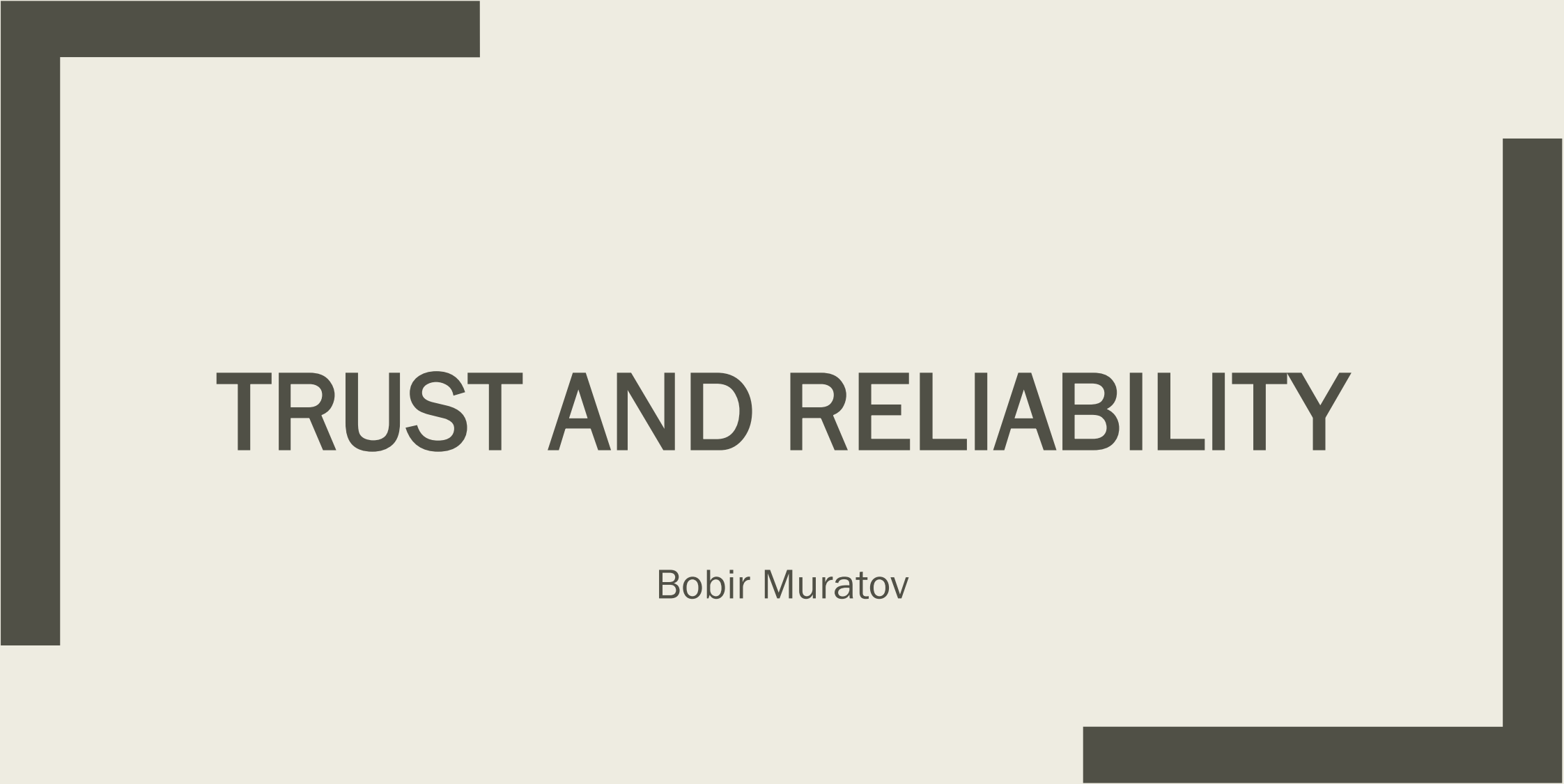


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TRUST AND RELIABILITY

Bobir Muratov

Case



Introduction

- Society has become dependent on professionals whose knowledge and expertise are not widely shared or understood
- “knowledge explosion”
- “ignorance explosion”

Honesty

- Honesty is a virtue
- Engineers are required:
 - “to be honest and realistic in stating claims or estimates based on available data.”
 - “to seek, accept, and offer honest criticism of technical work.”

Forms Of Dishonesty

■ Lying

- A lie ordinarily involves something that is believed by the liar to be false or seriously misleading
- A lie is ordinarily stated in words
- A lie is made with the intention to deceive

Forms Of Dishonesty

- Deliberate Deception
- Withholding Information:
 - if one fails to convey information that the audience would reasonably expect would not be omitted
 - if the intent of the omission is to deceive
- Failure to Seek Out the Truth

Why is Dishonesty Wrong?

Why is Dishonesty Wrong?

- Honesty as a Virtue:
- Norm Lewis case
- Is maintaining academic trustworthiness in some areas is more important than in others?

Why is Dishonesty Wrong?

- Dishonesty and Respect for Persons
- Moral agents
- Engineers have responsibility to ensure that employers, clients, and the general public can make autonomous decisions with understanding, particularly understanding of their consequences

Why is Dishonesty Wrong?

- **Utilitarian Considerations**

- Engineering contributes to utilitarian goals and informed decision-making

- **Trust and Truthfulness**

- “Whatever matters to human beings, trust is the atmosphere in which it thrives”

Dishonesty on Campus

- Problems for engineering students regarding honesty in engineering can arise even before completing their undergraduate work
- Case: would you report your friend's mistake?
- Case: capstone project

Dishonesty in Research and Testing

- What types of misconducts in research do you know?

Types of Dishonesty in Research and Testing

- Falsification of data
- Fabrication of data
- Failure to respect intellectual property of others
- Plagiarism
- Inappropriate attribution of authorship

Research Integrity Indicator

- <https://www.scimagoiris.com/>

Other Points of Trust and Reliability

- Confidentiality
- Expert Witnessing
- Informing the Public
- Conflicts of Interest?

Conflicts of Interest

- A conflict between an obligation to exercise good judgment and interest(s) that may compromise that judgment
- A conflict of interest is not just any set of conflicting interests
- Having more commitments than one can satisfy in a given period of time is not a conflict of interest
- The interests of the client, employer, or public that the engineer must protect are restricted to those that are morally legitimate
- A distinction is sometimes made between actual and potential conflicts of interest
- Although it is best to avoid conflicts of interest, sometimes this cannot reasonably be done

Midterm

- Chapter 1-5 (mostly from lectures)
- Multiple choice: 10
- Essay type: 10
- Concepts, definitions, applications-cases
- 60 minutes